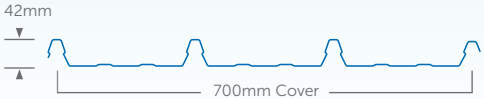
ROOFING AND WALLING

Roofing
Walling
Verandah
Fencing

TOPDEK® 700

Topdek 700 is a roof decking that is attached with a concealed clip. This ingenious system provides a clean and strong finish, without visible fixings. It allows for thermal expansion and contraction, making it ideal for long length roofing in both domestic and commercial uses.



Tolerance L ±5mm W ±2mm
Minimum Pitch 1°

MATERIAL SPECIFICATIONS

Material Properties	0.42mm BMT		0.48mm BMT	
	Zinc/Al	Colour	Zinc/Al	Colour
Total Coated Thickness (mm)	0.47	0.50	0.53	0.56
Mass (kg/linear metre)	3.26	3.32	3.70	3.76
Mass (kg/square metre)	4.66	4.74	5.28	5.37
Yield (square metre tonne)	214.6	211.0	189.4	186.2
Tensile Strength (MPa)	G550	G550	G550	G550

MAXIMUM RECOMMENDED SPANS (mm)

DETERMINED BY WIND SPEEDS FOR NON-CYCLONIC AREAS

Span Type	Roofing BMT		Walling BMT	
	0.42mm	0.48mm	0.42mm	0.48mm
Single Span	2000	2200	2850	3000
End Span	2200	2950	3000	3250
Internal Span	2300	3100	3200	3600
Un-stiffened Overhang	200	200	300	300
Stiffened Overhang	600	600	400	400

Roofing : Spans are limited based on typical maintenance foot traffic.
Walling : Spans based on N1 (W28) wind loading.
All spans are based on three fasteners per clip per support.

Domestic Verandah and Carport	Wind Classification			
	N1 (W28)	N2 (W33)	N3 (W41)	N4 (W50)
0.42mm BMT	3800	3800	2300	1500
0.48mm BMT	4200	4200	2600	1750

NOTE: Carport and Verandah spans are not based on fixing to clips, but on pierce fixing roof sheets with a minimum two fasteners per pan per supporting member. Contact Stratco if further advice is required.

TOPDEK 700 SPANS (mm)

DETERMINED BY WIND SPEEDS FOR NON-CYCLONIC AREAS

BMT	Application	Span Type	SPAN - THREE FASTENERS				SPAN - FIVE FASTENERS			
			N1 (W28)	N2 (W33)	N3 (W41)	N4 (W50)	N1 (W28)	N2 (W33)	N3 (W41)	N4 (W50)
0.42mm	Roofing	Single	2000	1750	1250	-	2000	1900	1500	-
		End	2200	2100	1550	-	2200	2200	1750	1300
		Internal	2300	2300	1650	-	2300	2300	2050	1550
	Walling	Single	2850	2050	1750	900	2850	2250	1900	1200
		End	3000	2400	1950	1350	3200	2650	2250	1850
		Internal	3200	2900	2250	1650	3600	3100	2550	2150
0.48mm	Roofing	Single	2200	1900	1600	-	2200	1900	1600	-
		End	2950	2300	2000	1300	2950	2300	2050	1550
		Internal	3100	2850	2300	1400	3100	2950	2400	2000
	Walling	Single	3000	2300	1900	1250	3000	2250	1900	1450
		End	3250	2750	2300	1800	3600	2850	2300	2050
		Internal	3600	3250	2850	2100	3600	3350	2950	2550

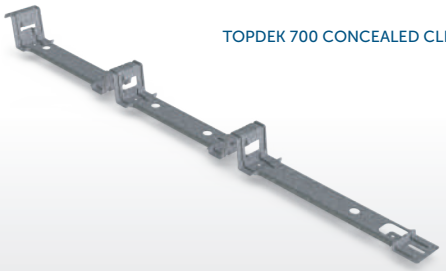
Values applicable for use with steel supports of minimum 0.95mm thickness, G550.

NOTE: For Topdek 700 span table wind capacities, design parameters and span definitions, please refer to the Cladding Wind Capacities tables and Engineering notes at the beginning of the Roofing and Walling section.

TOPDEK 700 CONCEALED CLIP

The most important feature of Topdek 700 is the single piece concealed clip. Its innovative design ensures maximum performance of the roofing, superior spanning and ease of installation. It is suitable for a variety of industrial and commercial uses, and can be used where a low roof pitch is required and security is important. Buildings such as shopping centres, hospitals, industrial storage, factories and warehouses lend themselves to this style of roofing for its strong, reliable, no fuss application.

- Specially designed single piece galvanised steel roof clip eliminates the possibility of mechanical failure.
- The clip is simply fastened onto the purlin and the roof deck locks straight onto the clip for fast and easy installation.
- The secure locking action ensures the deck is water tight and protects against entry.
- All fasteners are covered by the deck helping to minimise fixing corrosion.
- Specially designed recesses for fastening avoid denting the roof sheet.
- Allows for bracket adjustment to accommodate out of square buildings or installer requirements.



TOPDEK 700 CONCEALED CLIP

TOPDEK® 700 INSTALLATION

NOTE: The following recommendations apply to non-cyclonic areas.

STEP ONE: Align the top and bottom clips of the first sheet along the purlins with the arrow pointing in the laying direction, and fasten in positions 1, 2 and 3 (refer to Step 1) using the number and type of fasteners specified in the Topdek 700® span tables and fastener selection table. Run a string line between the two clips, or use the edge of the first deck to ensure all clips are in line. Mark the top, bottom and middle purlins at one metre from the starting line and at 700mm centres across the purlins. Use these marks to keep the sheets square.

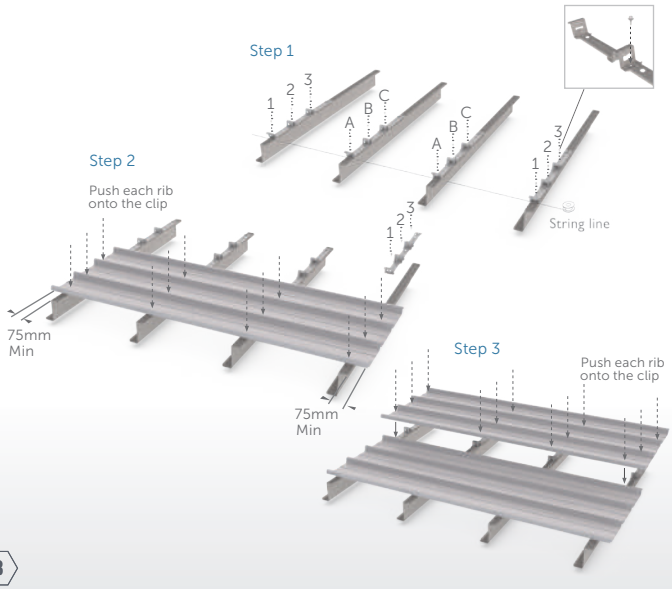
Align the remainder of the first run of clips to the string line and along the purlins. Fasten in position A, B and C (refer to Step 1) using the number and type of fasteners specified.

STEP TWO: Place the first sheet over the clips ensuring a minimum projection of 75mm at the ridge and into the gutter. Snap each rib onto the clip at every purlin using steady foot pressure, ensuring complete deck engagement at every rib on every clip, see step 2.

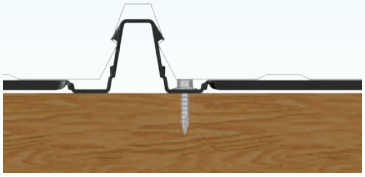
Hook the next run of clips over the last rib of the first sheet. The clips will align themselves and must be butted together for correct positioning. Align the hole with the slot and fasten at the joint first, then work along the bracket. During laying, regularly check the sheets for fanning or creep. If adjustment is necessary, the clips can be pulled away from the sheet while fastening.

STEP THREE: Place the second sheet over the clips, ensure the ends of the sheets align with the first sheet. Snap the deck onto the clips in the laying order using steady foot pressure, fully engage the interlocking rib onto each clip. It is important to ensure the deck is engaged on each rib, on every clip, and the deck is engaged along its entire length.

Install the remaining clips and sheets. Remember to frequently check the coverage to maintain squareness and adjust the creep where necessary. At the end of the purlin, cut the clips and deck where necessary to suit. To secure the final sheet edge, cut the end off the bracket, hook over the last rib and fasten. Finally, remember to turn up the pans at the ridge and down at the gutter end.



TOPDEK 700 FIXING DIRECTIONS - NON-CYCLONIC AREAS

	Roofing Concealed Clip Fixing
Fixing to Steel Hex head self drilling screw	
Fixing to Timber Type 17 self drilling wood screw	

TOPDEK 700 FASTENER SELECTION FOR NON-CYCLONIC AREAS

Support	Support Thickness	Insulation Blanket	Type of Fastener	Fastener Size
Steel	≤ 2.5mm	No	Hex head self drilling screw	No. 10 - 16 x 16mm
		Yes	Hex head self drilling screw	No. 10 - 16 x 25mm
	> 2.5mm ≤ 5.0mm	No	Hex head self drilling screw	No. 10 - 24 x 16mm
		Yes	Hex head self drilling screw	No. 10 - 24 x 25mm
	> 5.0mm	No	Drill 4.5mm tapping hole	No. 10 - 24 x 25mm
		Yes	Drill 4.5mm tapping hole	No. 10 - 24 x 25mm
Timber Hardwood	> 100mm	No or Yes	Type 17 self drilling wood screws	No. 10 - 12 x 30mm
Softwood	> 100mm	No or Yes	Type 17 self drilling wood screws	No. 10 - 12 x 30mm

NOTE:

- All fasteners must comply with Australian standard AS3566.1,2, Class 3.
- In the fastener screw sizes quoted above, the first number equals the screw gauge; the second number is the threads per inch and the third number is the screw length.
- Self drilling and tapping screws should NOT use neoprene washers when securing clips to purlins.
- The minimum number of screws along the clip is three, one at each end and one centrally.
- For additional strength, five screws may be used.
- If fixing over an insulation blanket the next standard screw length to that indicated may be required with minimum 25mm timber embedment to be maintained.